Parkinson's Disease Detection Using Handwriting

Abstract

Parkinson's disease is a progressive neurological disorder that affects motor control and handwriting ability. Early detection of Parkinson's disease is important for timely medical treatment and monitoring. In this research, a deep learning based approach is proposed to detect Parkinson's disease using handwriting images. The system analyzes spiral and wave drawings using a Convolutional Neural Network (CNN) model. The HandPD dataset is used for training and testing the model. The dataset contains handwriting images from healthy individuals and Parkinson's disease patients. Experimental results show that the proposed model achieves an overall accuracy of approximately 87.93%. The results demonstrate that handwriting analysis combined with deep learning techniques can be an effective tool for early Parkinson's disease detection.

Keywords

Parkinson's Disease, Handwriting Analysis, Deep Learning, CNN, Medical Image Classification

1. Introduction

Parkinson's disease is a neurodegenerative disorder that primarily affects movement and motor control. Patients suffering from Parkinson's disease often experience tremors, rigidity, and difficulties in performing fine motor tasks such as handwriting. Handwriting abnormalities are considered one of the early indicators of Parkinson's disease.

Traditional diagnosis methods rely on clinical observations by neurologists. However, automated systems based on artificial intelligence can assist doctors in detecting the disease more efficiently. Recent advancements in deep learning have shown significant improvements in medical image analysis and pattern recognition.

2. Literature Review

Several studies have explored the use of handwriting analysis for Parkinson's disease detection. Pereira et al. proposed a method for analyzing handwritten dynamics using convolutional neural networks to identify Parkinson's disease symptoms [1] .

Spiral drawing analysis has also been used for objective assessment of motor symptoms in Parkinson's patients [2] . Deep learning techniques have become widely used in medical image classification tasks because they automatically extract complex features from images [3] . CNN models have achieved great success in image recognition and classification tasks [4] .

3. Dataset Description

HandPD Dataset

The dataset used in this research is the **HandPD dataset**, which contains handwriting exams collected from both healthy individuals and patients diagnosed with Parkinson's disease. The dataset was collected at the **Botucatu Medical School, São Paulo State University (UNESP), Brazil.** It consists of spiral and meander (wave) drawings used to analyze handwriting patterns affected by Parkinson's disease.

Dataset Source

The dataset can be accessed from the official research page:

HandPD Dataset:

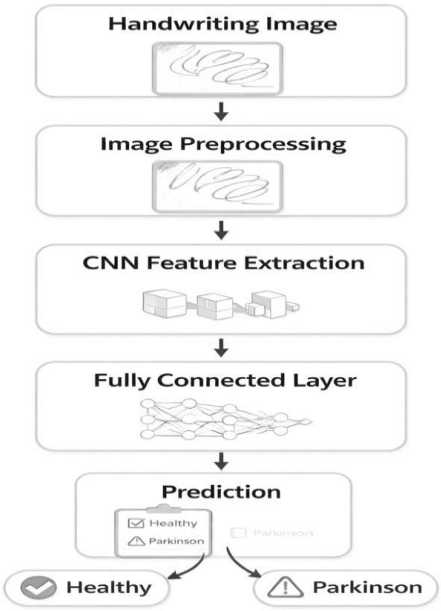
https://wwwp.fc.unesp.br/~papa/pub/datasets/Handpd/

Feature	Description
Dataset Name	HandPD
Total Participants	92
Healthy Subjects	18
Parkinson Patients	74
Total Images	~736
Image Type	Spiral and Wave Drawings

4. Proposed System

The proposed system detects Parkinson's disease using handwriting images. The system includes image preprocessing, CNN based feature extraction, and classification.

System Workflow:



5. Methodology

All handwriting images are resized to **128×128 pixels** and converted to grayscale. Pixel values are normalized between **0 and 1** to improve model performance.

Data augmentation techniques are also applied to increase dataset diversity and reduce overfitting. These techniques include:

- Rotation
- Width shift
- Height shift
- Zoom

These preprocessing steps help the model learn better features from handwriting images.

7. Experimental Results

Dataset Type	Accuracy
Spiral Drawings	80.00%
Wave Drawings	94.68%
Overall Model	87.93%

The results show that the proposed system performs well in detecting Parkinson's disease using handwriting images.

8. Conclusion

In this research, a CNN based deep learning model was developed for detecting Parkinson's disease using handwriting analysis. Spiral and wave handwriting images from the HandPD dataset were used for training and testing. The proposed system achieved an overall accuracy of approximately 88%, demonstrating the effectiveness of deep learning techniques in medical diagnosis.

9. Future Work

Future improvements may include increasing the dataset size, applying advanced deep learning architectures, and developing mobile or web-based applications for real-time Parkinson's disease detection.

References.

[1] Pereira C.R., Weber S.A., Hook C., Papa J.P. Handwritten Dynamics Assessment through CNN for Parkinson's Disease Identification. Artificial Intelligence in Medicine, 2016.
Paper: A new computer vision-based approach to aid the diagnosis of Parkinson's disease - ScienceDirect

[2] Bala D., et al. Spiral Analysis for Objective Assessment of Motor Symptoms in Parkinson's Disease. IEEE Transactions on Biomedical Engineering, 2010.
Paper: Automatic Spiral Analysis for Objective Assessment of Motor Symptoms in Parkinson's Disease - PubMed

[3] LeCun Y., Bengio Y., Hinton G. Deep Learning. Nature, 2015.
Paper: Deep learning | Nature

[4] Krizhevsky A., Sutskever I., Hinton G. ImageNet Classification with Deep Convolutional Neural Networks. NIPS, 2012.
Paper: ImageNet Classification with Deep Convolutional Neural Networks